

[Kat]herine Titterton

San Francisco, CA · [macrofluidic.me](#) · X @KatTitterton

TL;DR Interdisciplinary shamanka working at the seam of informatics, software, and hardware in biotech. 8+ years turning bench work into machine-learning fuel: building and scaling high-content / high-throughput assays, then the computational image analysis that unlocks phenotypic discovery at absurd throughput. Equal parts process engineer, artist in residence, taste engineer, and shameless personality hire. I find the shortest path and walk it.



[MACROFLUIDIC.ME/RESUME](#)

EXPERIENCE

● Paper / preprint ● Poster

Founding Member of the Technical Staff, Axion Bio · [axi.om](#)

2023–Present · San Francisco, CA

Developed advanced new approach methodologies (NAMs) enabling medicinal chemists to understand the bioactivity profile of their molecules and design better, safer drugs. Wore many hats: feature engineering, algorithm development, and scientific product / pivot-direction steering. Oversaw assay development and data generation for two of the world's largest Cell Painting datasets. Invented two unique bioactivity (morphology-change) detection methods that sensitively detect small-molecule effects at ~10× lower concentrations than conventional toxicity assays (LDH, MT-Glo, cell count). Honorary Artist in Residence. Keeper of poison dart frogs. (Ask me about the last one!)

- Seal S, Dee W, Shah A, Zhang A, **Titterton K**, Cabrera AA, et al. Small molecule bioactivity benchmarks are often well-predicted by counting cells. *bioRxiv* 2025.04.27.650853 (2025).
- Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *bioRxiv* (2025). PMID: PMC11785178.
- **Titterton K**, Boiko D, Cabrera AA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure. *Society of Toxicology (SOT) Annual Meeting*, 2025. [macrofluidic.me/13558658-2025-sot](#)

Associate Scientist III, Insitro · Arrayed Screening, Neuroscience

2022–2023 · South San Francisco, CA

Adapted high-content phenotypic assays for high-throughput small-molecule screening (tuberous sclerosis / TSC and ALS), scaling onto near-fully-automated cell feeding, fixation, ICC, and imaging platforms (BioTek, High-Res). Developed a digital phase-contrast timecourse imaging modality and corresponding Python analysis suite (pre-LLM) to track neuron growth dynamics and drive label-free phenotypic discovery.

Technical Associate II, Massachusetts Institute of Technology · Synthetic Neurobiology Group

2021–2022 · Cambridge, MA

Optimized multiplexed expansion revealing (mExR) stripping technique for 7+ rounds of IHC on 16× linearly-expanded specimens, enabling simultaneous nanoscale visualization of 30+ proteins. Built and routinely executed image-analysis scripts in ImageJ/Fiji and MATLAB.

- Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, et al. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun* 15(1):9722 (2024). PMID: PMC11550395.

Associate Scientist II, AbbVie · Foundational Neuroscience Center, Early R&D

2018–2021 · Cambridge, MA

Specialized in primary murine and hiPSC neuronal tau-aggregation high-content analysis / screening (HCA) using the Revvity Opera confocal system and Harmony software. Used microfluidic devices to model intercellular propagation of tau pathophysiology in AD. Wrote R scripts for automated, reproducible dose-response generation to profile and rank lead candidates.

- Wu JW, **Titterton K**, Sebastian R, Romanul N, et al. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem* (2025). PMID: 41135680.

- Tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation. *SfN Global Connectome*, Jan 2021. [macrofluidic.me/9001434-2021-sfn-poster](#)
- HCA of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro. *SBI2 High Content Meeting*, Sep 2019, Harvard Medical School. [macrofluidic.me/9001442-2019-sbi2-poster](#)

Research Technician I, Massachusetts General Hospital

2017–2018 · Boston, MA

Used voltage-clamp electrophysiology in GABAR-expressing *Xenopus* oocytes to elucidate mechanisms of general anesthetics.

- Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, et al. Inhibitable photolabeling by neurosteroid diazirine analog in the β 3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem* 162:810–824 (2019). PMID: PMC6912599.

PROJECTS

Bare Metal

2025

First IRL art exhibit at Human Tech week, SF, Jun 2025. Post-hoc analysis on [macrofluidic.me/blog](#).

Vivarium 2.0

2024

A lavish home for Axion Bio's pet poison dart frogs. Didn't get it right the first time; rebuilt from scratch, now with an automated water-misting system and a motion-triggered camera that captures live video recordings of the frogs.

OpenTransformers (NYC)

2023

A companion automation engineer for wet-lab scientists: unlocking the OT-2 robot and Python SDK via LLM code generation. 2nd place at BioHack NYC 2023.

EDUCATION

Harvard University · Extension School, graduate coursework

2017–2020

Completed while employed full-time: Social Medicine; Intro to Programming (Python); Image Processing & Computer Vision (Python).

B.A., Colgate University · *cum laude*

2013–2017

Major: Cellular Neuroscience. Minor: Mathematical Systems Biology.

Coursework: Systems Biology, Bioinformatics, Biostatistics, Mathematical Biology (Differential Equations), Biophysics, Neuroethology, Epidemiology, Bioethics, Psychopharmacology, Organic Chemistry, Biochemistry, Multivariable Calculus, Linear Algebra. Three summer-research posters (Huntington's RNAi in transgenic zebra finch; atopy & helminth infection in Ethiopian schoolchildren; PDE2 regulation of β -arrestin signaling, Mount Sinai). Co-captain, Division I Track & Field (2016–17); Dorothy-Waltham Award for Athletic Leadership. Marine Mammal Cognition extended study, Dolphin Research Center, FL.

SKILLS & TOOLS

COMPUTATIONAL

LLM codegen (Claude Code, Cursor) · image embeddings (visualization, clustering, feature derivation, normalization) · Python (data science, image processing & segmentation; Jupyter, NumPy, SciPy, scikit-learn, pandas, matplotlib, seaborn, xlsxwriter, Colab) · R (ggplot2, tidyverse, drc, platetools) · dose-response curve fitting (GraphPad Prism, R, Python) · uni/multivariate statistics · design of experiments (DOE) · ImageJ/Fiji macro scripting · Benchling ELN.

WET LAB

Cell culture (primary murine neuron, hiPSC neuron models) · microfluidic cell-culture models · ICC / IHC · confocal microscopy (manual, PE Operetta / Opera Phenix) · high-content image analysis (PerkinElmer Harmony) · automated liquid handling & schedulers (OpenTrons, Integra Assist Plus / Viaflo, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go) · assay development & HT/HC screening · stereotaxic surgery / tissue microdissection · voltage-clamp electrophysiology.

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)